

BUKOKA, Tanzania, United Republic of

WMO: 637290

Lat: 1.3328S

Lon: 31.8214E

Elev: 3730

StdP: 12.82

Time Zone: 3.00 (E03)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	62.3	62.9	55.1	74.3	68.1	56.8	78.9	68.1	16.6	75.6	12.0	74.0	1.0	240	0.209

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.5	82.8	71.2	82.3	71.1	81.0	70.7	73.5	80.8	72.7	80.0	72.0	79.4	6.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	131.6	78.7	70.1	127.4	77.8	69.7	125.6	77.3	39.7	81.0	39.0	80.2	38.4	79.3	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	12.1	10.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			55.7	77.3	1.8	1.6	54.3	78.5	53.3	79.4	52.3	80.2	50.9	81.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.0	72.5	73.2	72.6	72.2	71.8	71.9	71.2	71.4	71.6	72.2	71.9	71.8
	DBStd	1.70	1.83	1.96	2.01	1.57	1.47	1.31	1.30	1.32	1.50	1.50	1.65	1.74
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	8032	696	650	701	665	677	656	656	663	648	687	658	675
	CDD65	2557	231	230	236	215	212	206	191	198	198	222	208	210
	CDH74	10150	1079	1239	1090	755	710	834	757	742	737	804	696	707
	CDH80	549	102	160	117	25	7	10	10	11	23	30	21	33
(14) Wind	WSAvg	2.6	1.8	2.4	2.3	2.3	2.9	3.5	3.5	3.3	2.7	2.5	2.0	1.7
(15) Precipitation	PrecAvg	84.3	6.1	6.6	10.0	14.8	12.6	3.2	2.0	2.8	3.9	6.6	8.1	7.7
	PrecMax	108.3	11.3	12.4	16.1	24.8	20.6	10.1	7.2	5.6	8.7	14.6	18.8	14.3
	PrecMin	65.4	1.9	1.9	3.4	4.6	5.6	0.2	0.0	0.3	1.5	3.1	4.1	1.9
	PrecStd	10.7	2.4	3.1	2.3	4.2	3.9	1.9	2.0	1.4	1.9	3.1	3.3	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.0	84.5	84.4	82.2	80.8	80.9	80.9	81.0	82.0	82.4	82.0	82.5
		MCWB	71.5	71.3	72.3	71.8	70.7	69.3	69.0	70.2	70.4	70.1	70.3	69.5
	2%	DB	82.4	82.8	82.7	80.6	79.5	80.1	79.8	80.1	80.6	80.8	80.5	80.8
		MCWB	71.2	71.4	71.7	71.5	70.4	69.4	68.4	69.8	70.4	70.6	70.6	70.2
	5%	DB	80.9	82.2	81.2	79.2	78.8	79.0	78.9	78.9	79.2	79.6	79.1	79.6
		MCWB	70.8	71.4	71.3	70.9	70.2	69.0	68.0	69.0	69.7	70.1	70.2	70.1
	10%	DB	79.8	80.7	80.3	78.4	77.8	78.4	77.6	77.8	78.5	78.6	78.3	78.4
		MCWB	70.4	71.0	70.9	70.6	69.8	68.7	67.4	68.3	69.4	69.7	70.0	69.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.1	74.1	74.2	73.3	72.8	72.0	71.8	72.9	72.8	74.1	72.8	72.7
		MCDB	81.5	82.0	82.0	80.1	78.9	78.7	78.6	79.0	79.8	79.5	79.7	79.4
	2%	WB	72.6	72.9	72.8	72.3	71.8	70.9	70.4	71.0	71.6	71.9	71.7	71.5
		MCDB	80.3	81.0	81.0	79.1	78.2	78.0	77.5	78.3	79.0	78.9	78.7	78.7
	5%	WB	71.7	72.0	71.9	71.6	71.0	70.0	69.5	70.0	70.6	70.8	70.9	70.8
		MCDB	79.4	80.2	80.1	78.3	77.3	77.5	76.8	77.3	77.9	77.9	77.8	77.7
	10%	WB	70.9	71.3	71.1	70.8	70.3	69.2	68.7	69.2	69.8	70.0	70.1	70.0
		MCDB	78.2	79.4	78.9	77.2	76.5	76.6	76.0	76.3	77.0	77.1	76.9	76.7
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	14.5	13.8	11.9	11.7	12.9	14.0	13.3	13.5	12.9	12.7	12.7
		MCDBR	16.5	16.8	16.3	13.7	13.3	14.3	15.7	15.1	15.6	14.7	14.4	14.8
	5% WB	MCWBR	9.5	10.2	9.9	7.6	7.4	7.8	8.5	8.7	9.1	8.1	7.7	8.1
		MCWBR	15.4	15.7	15.2	12.9	12.7	13.4	14.0	14.1	14.8	13.7	13.7	14.0
(40) Clear-Sky Solar Irradiance	taub		0.436	0.437	0.411	0.362	0.360	0.396	0.427	0.432	0.395	0.384	0.373	0.408
		taud	2.306	2.312	2.430	2.597	2.591	2.436	2.304	2.297	2.416	2.453	2.517	2.404
	Ebn at Noon		284	285	290	297	291	275	268	273	290	297	302	292
		Edh at Noon	43	43	38	32	30	35	40	42	38	37	35	39
(44) All-Sky Solar Radiation	RadAvg		1545	1701	1616	1551	1531	1604	1651	1591	1552	1531	1427	1423
	RadStd		162	104	121	137	93	143	113	126	110	110	90	123

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page